

Final Report

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1. Abstract

2. Introduction

Total fertility rates (TFRs) are key population indicators that are widely monitored because they influence age structure and population growth (Wilmoth et al.2022; World Health Organisation). Globally, fertility rates have declined over recent decades (Chen et al. 2018; Shair et al. 2024; Vanella 2024). In Singapore, the total fertility rate has dropped below the replacement level required for a population to replace itself across generations (Jones & Hamid 2015; Tan, 2024; Wilmoth et al. 2022). Therefore, forecasting TFR and total live births (TLB) may provide insight into future fertility patterns and population change in Singapore. Preliminary exploratory data analysis indicated a long-term decline in TFR and TLB with periodic fluctuations, motivating the use of forecasting models capable of capturing these patterns.

Data from 1960 to 2024 on TLB and TFR were obtained from the Singapore Department of Statistics (2023). The research question addressed in this report is: Can ARIMA models accurately forecast Singapore's TLB and TFR while capturing short-term fluctuations?

This report is structured as follows. Part 3 describes the fertility decline in Singapore, Part 4 describes the process and workflow of obtaining viable ARIMA models,

3. Literature Review

Around 1965, Singapore introduced policies aimed at reducing population growth, which contributed to subsequent declines in fertility rates (Chen et al. 2018; Goodkind 1996; Singh et al. 1988; Vanella 2024). Despite this long-term decline, there may be cyclical patterns in TFR and TLB. These fluctuations may be associated with cultural preferences surrounding Lunar zodiac Dragon years (1964, 1976, 1988, 2000, 2012 and 2024), which are considered especially auspicious years and have been associated with higher fertility rates and live births (Goodkind 1996; Jones & Hamid 2015). Forecasting such series therefore requires models that can capture both long-term trends and short-term variation. Among the various models for time series forecasting, ARIMA models are particularly flexible because they allow selection from a broad class of models rather than imposing a single model form on every time series (Kolambe & Arora 2024; Newbold 1983). Consequently, ARIMA models were selected as the forecasting approach for Singapore's TFR and TLB.

4. Methods

For both TFR and TLB, an identical modelling workflow was applied. The analysis focused on the period from 1960-2012 for model fitting, with observations 2013-2024 reserved for evaluating forecast accuracy. Initial plots were used to examine the presence of trends, seasonality, and other notable patterns, which indicated a long-term decreasing trend in both series. Several transformations were considered including logarithmic transformation and first- and second-order differencing. Stationarity was assessed using the KPSS test. For each transformed series, autocorrelation (ACF) and partial autocorrelation (PACF) plots were used to find candidate ARIMA models. The candidate ARIMA models were then fitted and the residual ACF and PACF were examined to assess whether the residuals resembled white noise. Forecast accuracy was evaluated using mean square error (MSE) and mean absolute error (MAE) while model selection was guided by Akaike's Information Criterion (AIC) and the corrected Akaike's Information Criterion (AICc). This procedure was applied consistently to both series to ensure comparability of results.

5. Results

Table 1. TFR - no log transformation

Model	AIC	AICc	MSE	MAE
ARIMA(16,1,0)	-63.99	-45.99	0.04753075	0.1690442
ARIMA(12,2,3)	-62.57	-46.57	0.01218546	0.08274547

Table 2. TFR - log transformed

Model	AIC	AICc	MSE	MAE
ARIMA(13,1,3)	-136	-118	0.02602064	0.121592
ARIMA(13,2,2)	-132.39	-116.39	0.007426324	0.06826933

Table 3. TLB - no log transformation

Model	AIC	AICc	MSE	MAE
ARIMA(13,1,1)	974.57	987.9	82856991	7698.072
ARIMA(13,1,2)	973.01	988.56	54382525	6014.802

Table 4. TLB - log transformed

Log Transformed TLB Model	AIC	AICc	MSE	MAE
ARIMA(15,1,0)	-145.56	-130.02	2239721240	47245.34
ARIMA(12,1,3)	-139.69	-124.15	1752591080	41851.92

6. Discussion

For TFR, the initial model, ARIMA(16,1,0) has the lowest AIC (-63.99) but its forecast accuracy was lower compared to the other models, including the log transformed models. Furthermore the KPSS test suggested the series remained stationary after first-order differencing. Hence models based on second-order differencing were considered and ARIMA(12,2,3) has the lowest AICc (-46.57) and achieved better forecast accuracy. This suggests that second-order differencing provided a better representation of the underlying series.

Among the log transformed data, ARIMA(13,1,3) produced the lowest AIC (-136) but the forecast accuracy were did not perform as well as ARIMA(13,2,2) which has the lowest AICc (-116.39) and lowest MSE (0.007426) and lowest MAE (0.06827) across all TFR models. These results suggest that applying a log transformation together with second-order differencing improved the forecast performance for the TFR series.

For TLB, the ARIMA(13,1,2) model has the lowest AIC (973.01) and produced lowest forecast errors than the other TLB models (MSE: 54,382,525 & MAE: 6,014.80). ARIMA(13,1,1) model has the lower AICc (987.9) but its forecast accuracy was lower.

Among the log-transformed models, ARIMA(15,1,0) model produced the lower AIC (-145.56) and AICc (-130.02) but lower forecast accuracy. While ARIMA(12,1,3) had higher AIC and AICc values, it had a better forecast accuracy than ARIMA(15,1,0), log-transformed models resulted in considerably larger forecast errors when transformed back to the original scale.

These results suggest that first-order differencing was sufficient to account for the long-term trend in TLB, while logarithmic transformation did not improve forecast accuracy. However, the selected model contains a large number of autoregressive and moving average terms. Furthermore, ARIMA models rely solely on historical observations and do not explicitly incorporate external factors such as policies, economic status, or cultural influences. Therefore, while the model performed reasonably well for the evaluation period, forecast accuracy may decline if future birth patterns differ substantially from historical trends.

7. Conclusion

8. References

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- Wilmoth, J., Menozzi, C. & Bassarsky, L. (2022). *Why population growth matters for sustainable development* (Policy Brief No. 130), United Nations Department of Economic and Social Affairs, Population Division. https://www.un.org/development/desa/pd/sites/www.un.org.development.desa.pd/files/undesapd_2022_policy_brief_population_growth.pdf
- World Health Organization. *Total fertility rate (per woman)*. Accessed June 7, 2026, from <https://www.who.int/data/gho/indicator-metadata-registry/imr-details/123>

9. Appendix

Figure 1. Initial Plot of Total Fertility Rate (TFR)

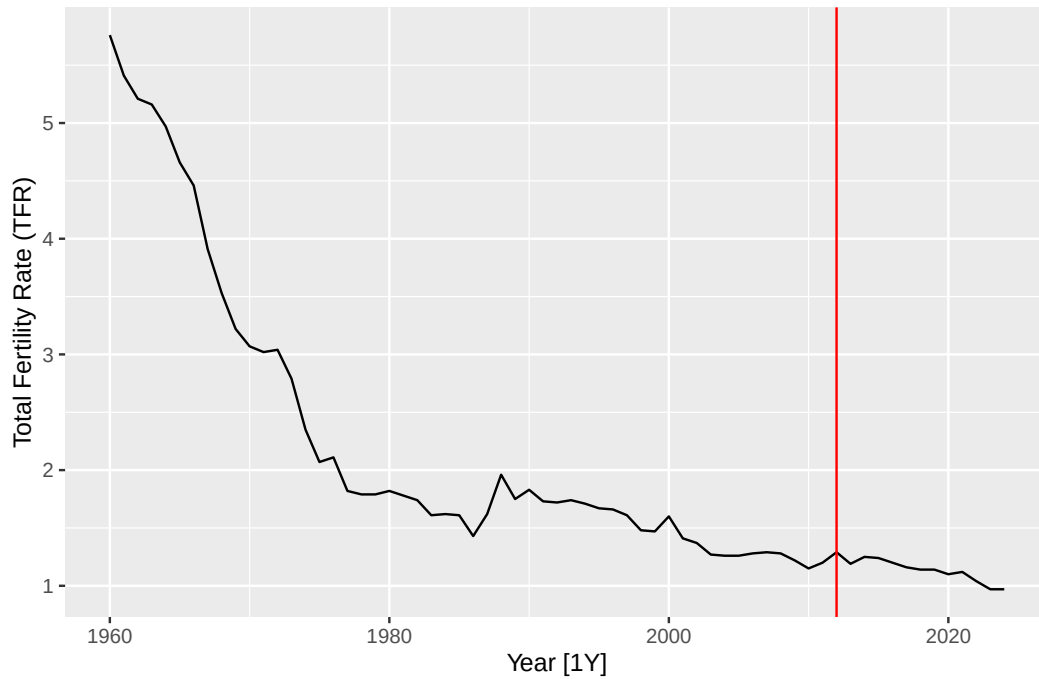
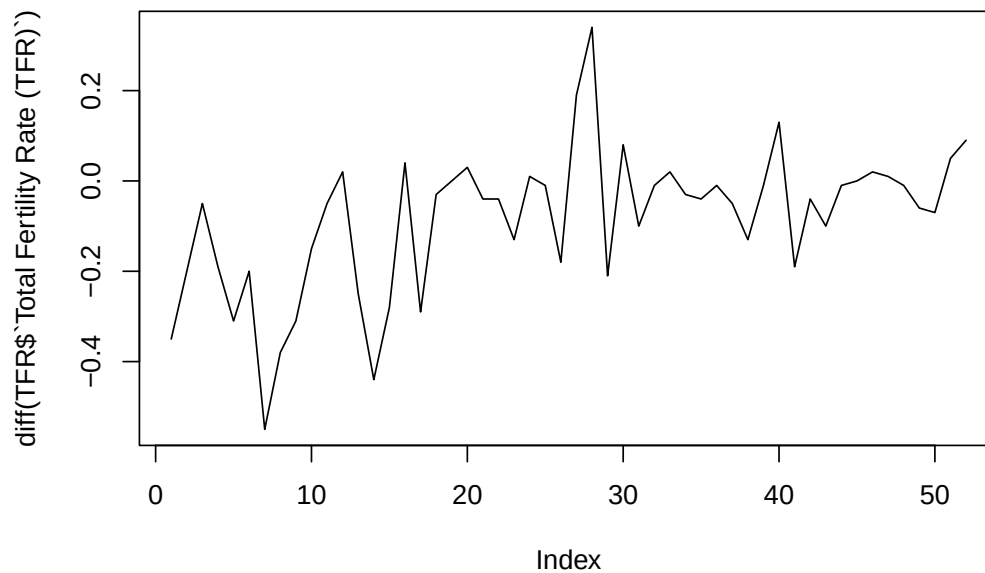
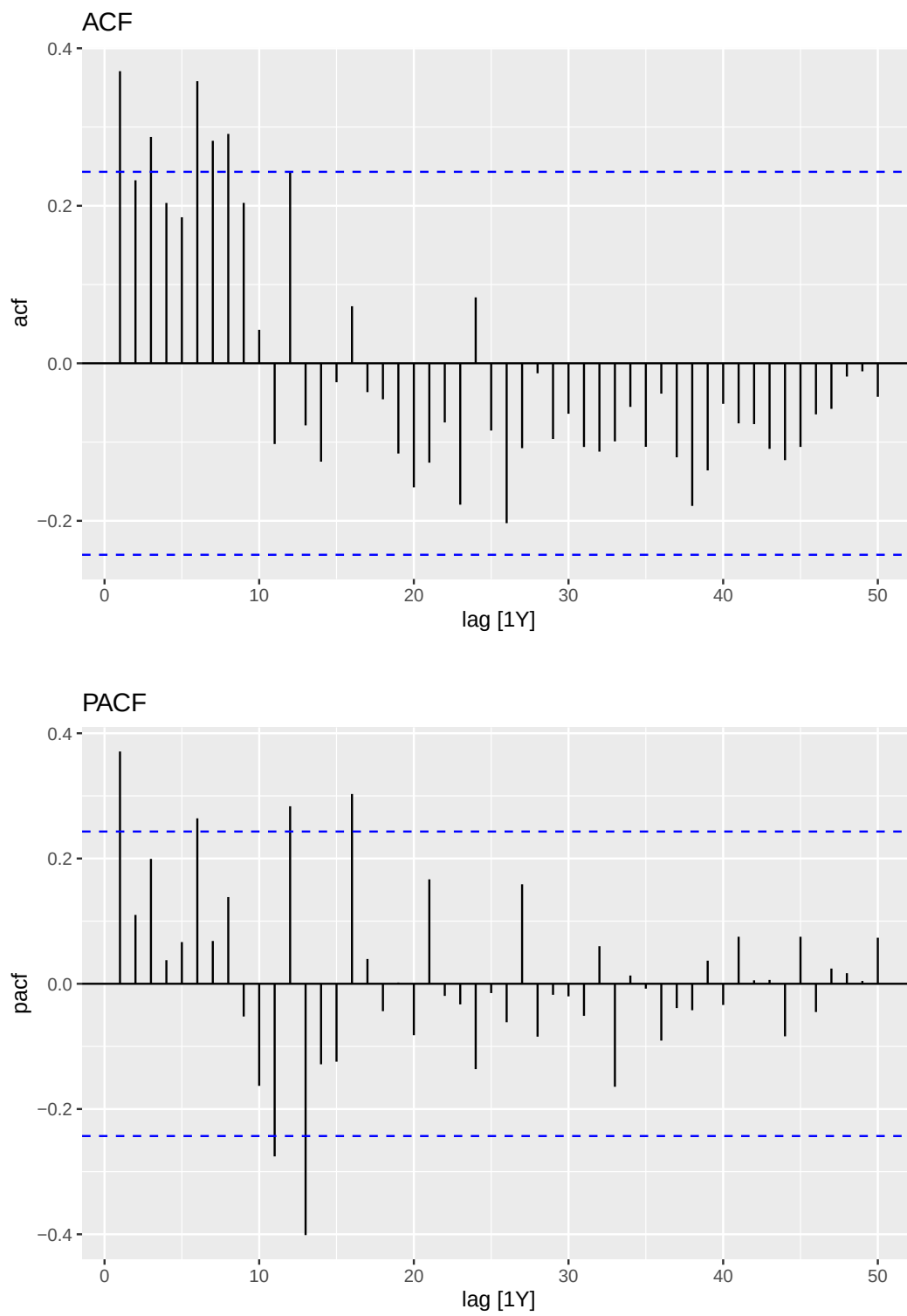


Figure 2. Plot of Differenced Total Fertility Rate (TFR) until 2012



```
##      kpss_stat kpss_pvalue
##      0.8412374  0.0100000
```

Figure 3. ACF and PACF Plots for Differenced Total Fertility Rate (TFR)



```

## Series: TFR_ts
## ARIMA(16,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.3647 -0.1061  0.4577 -0.2586  0.2984  0.2452  0.2006  0.1578 -0.0293
## s.e.  0.1240  0.1289  0.1246  0.1128  0.0935  0.0832  0.0952  0.0928  0.0996
##      ar10     ar11     ar12     ar13     ar14     ar15     ar16
##      -0.0306 -0.3594  0.3889 -0.6048 -0.0419 -0.3361  0.5135
## s.e.  0.0973  0.0913  0.1039  0.1066  0.1350  0.1295  0.1397
##
## sigma^2 = 0.009716: log likelihood = 48.99
## AIC=-63.99  AICc=-45.99  BIC=-30.82

```

Fitted ARIMA(16,1,0) model for TFR:

$$\begin{aligned}
& (1 - 0.3647B + 0.1061B^2 - 0.4577B^3 + 0.2586B^4 - 0.2984B^5 - 0.2452B^6 \\
& \quad - 0.2006B^7 - 0.1578B^8 + 0.0293B^9 + 0.0306B^{10} + 0.3594B^{11} \\
& \quad - 0.3889B^{12} + 0.6048B^{13} + 0.0419B^{14} + 0.3361B^{15} - 0.5135B^{16})(1 - B)y_t \\
& = c + \varepsilon_t
\end{aligned}$$

Figure 4. ACF and PACF plots of residuals for the ARIMA(16,1,0) model fitted to TFR

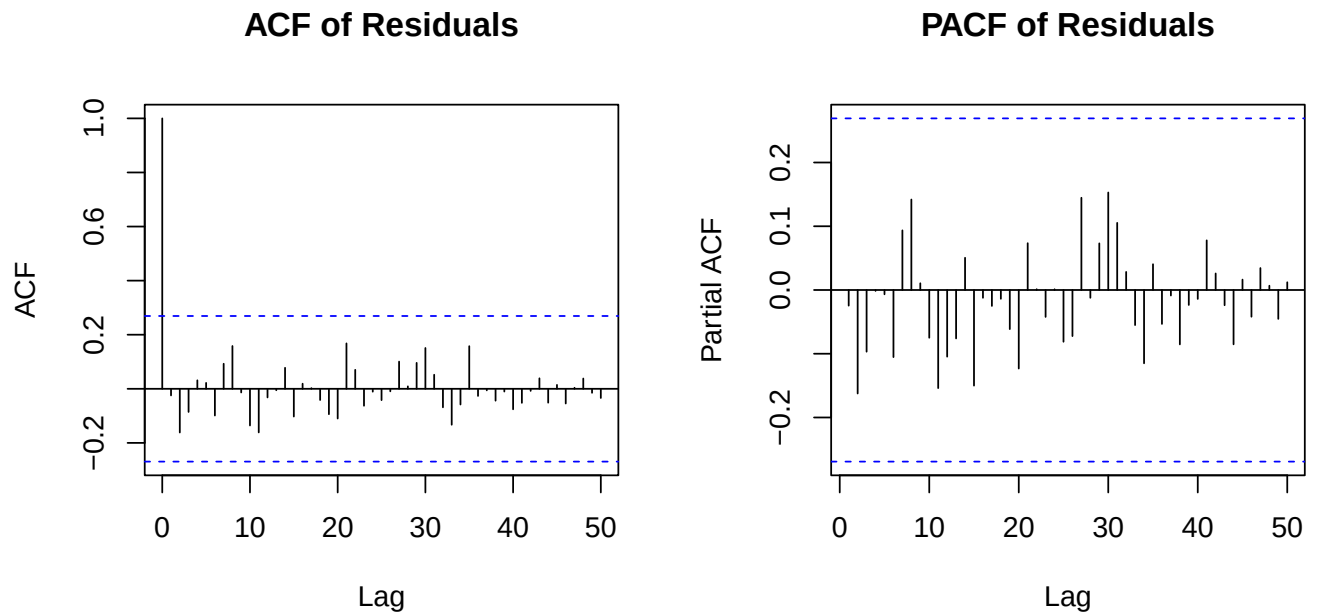
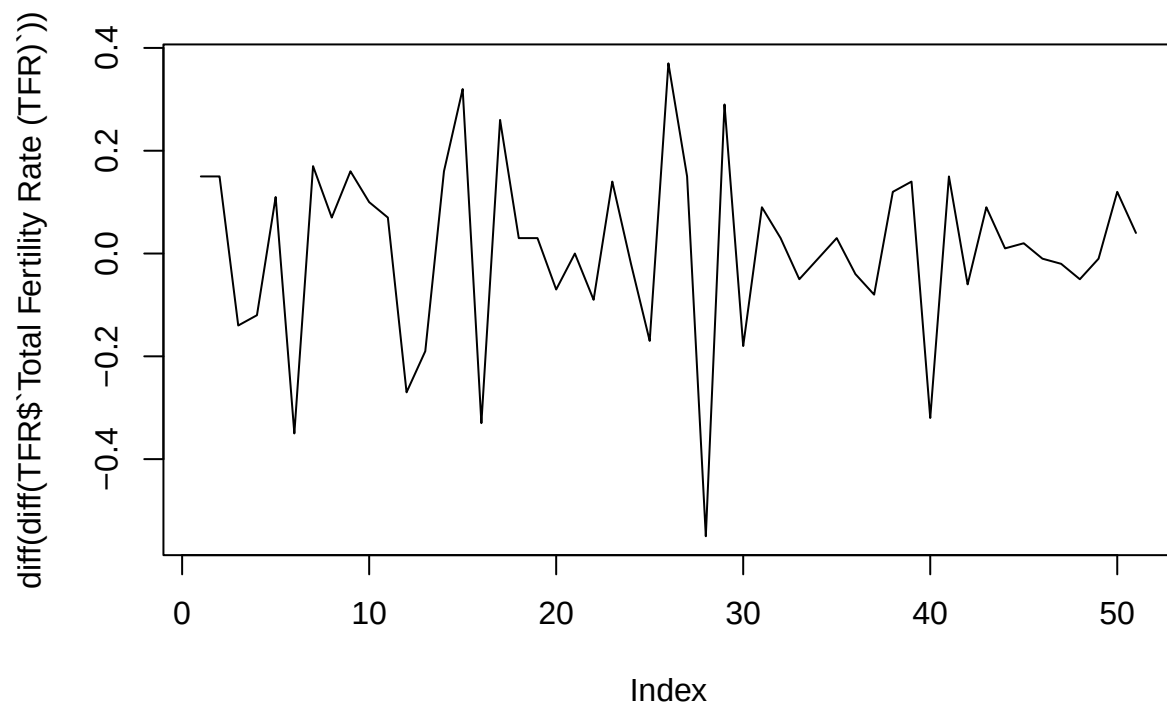
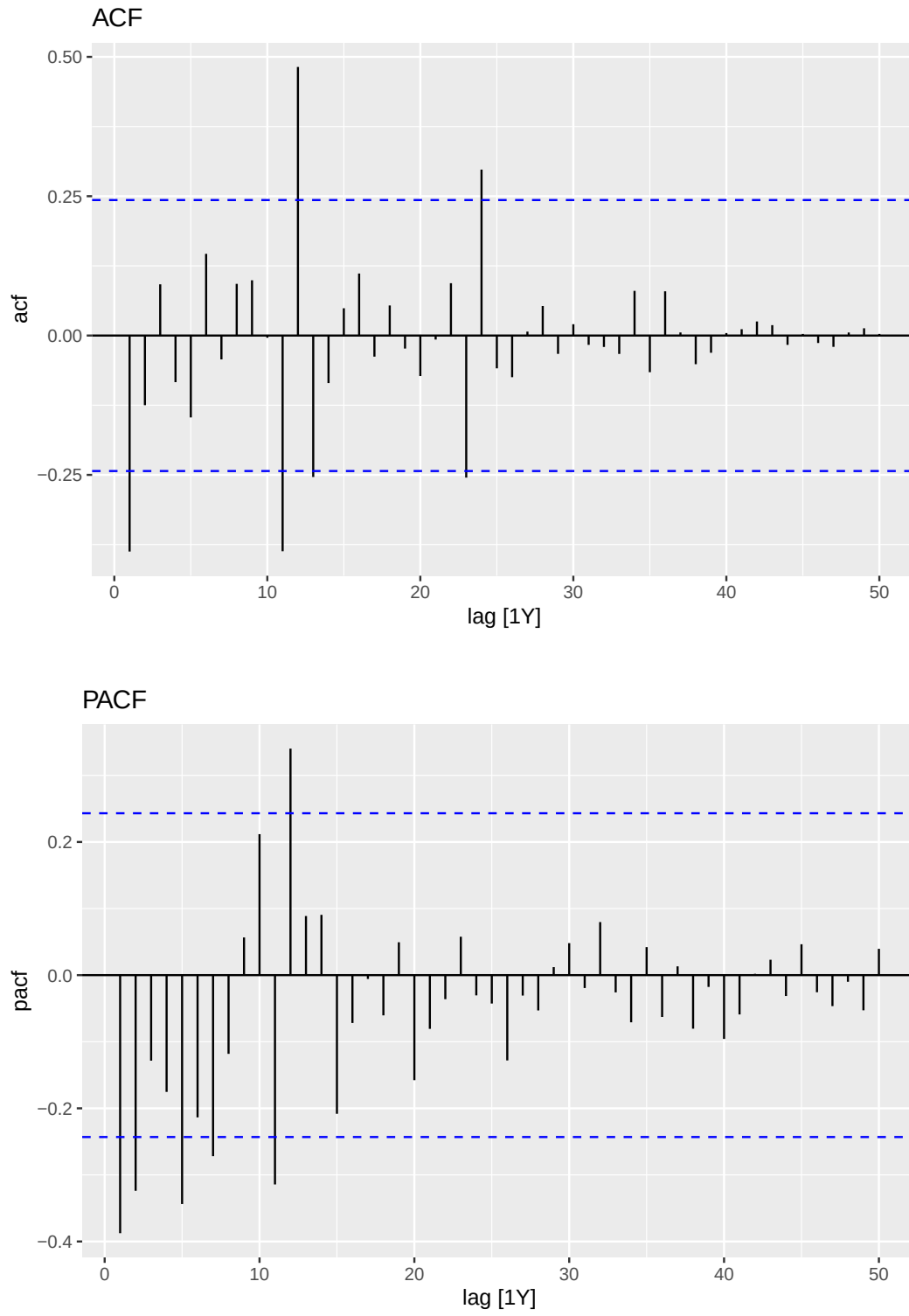


Figure 5. Plot of Differenced Twice Total Fertility Rate (TFR) until 201



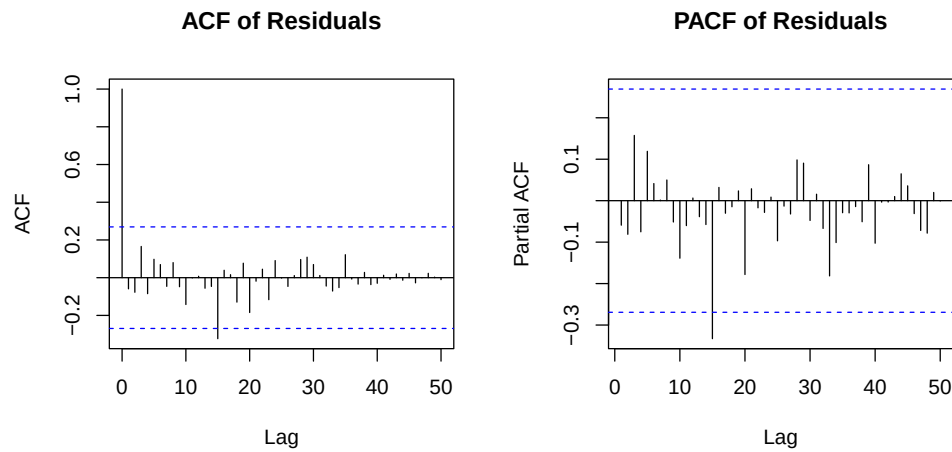
```
##      kpss_stat kpss_pvalue
## 0.04262877 0.10000000
```


Figure 6. ACF and PACF Plots for Twice Differenced Total Fertility Rate (TFR)



```
## Series: TFR_ts
## ARIMA(12,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.6234 -0.7441 -0.5813 -0.5700 -0.4163 -0.1741  0.0462  0.3482
## s.e.   0.1203   0.1494   0.1673   0.1676   0.1705   0.1772   0.1846   0.1847
##      ar9      ar10     ar11     ar12
##      0.4206   0.4880   0.1609   0.5040
## s.e.   0.1845   0.1783   0.1540   0.1205
##
## sigma^2 = 0.01286: log likelihood = 41.78
## AIC=-57.55   AICc=-47.72   BIC=-32.44
```

Figure 6. ACF and PACF plots of residuals for the ARIMA(12,2,0) model fitted to TFR



```

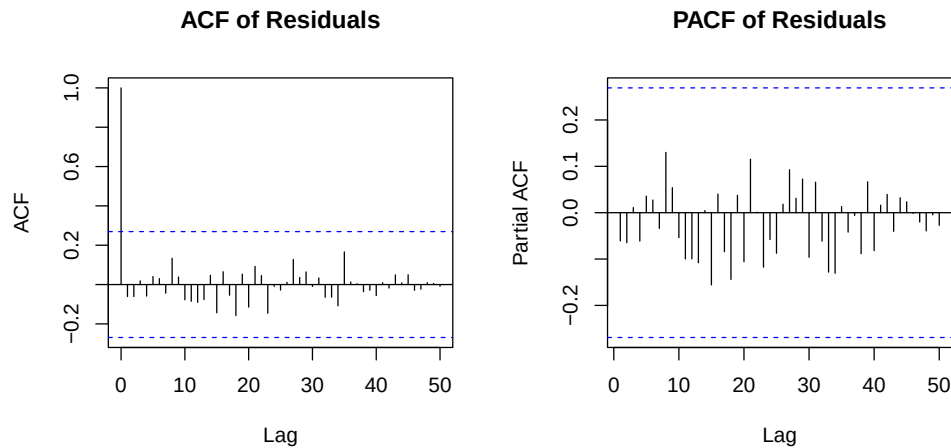
## Series: TFR_ts
## ARIMA(12,2,3)
##
## Coefficients:
##          ar1          ar2          ar3          ar4          ar5          ar6          ar7          ar8
##        -0.4622   -0.7606   -0.9273   -0.8808   -0.6967   -0.4124   -0.2312   0.0490
## s.e.    0.1091    0.1266    0.1643    0.2032    0.2376    0.2529    0.2548    0.2404
##          ar9          ar10         ar11         ar12         ma1          ma2          ma3
##          0.1578    0.3268    0.1426    0.6463   -0.1402    0.1411    0.8593
## s.e.    0.2067    0.1645    0.1248    0.1005    0.1237    0.1262    0.1309
##
## sigma^2 = 0.009277:  log likelihood = 47.29
## AIC=-62.57   AICc=-46.57   BIC=-31.66

```

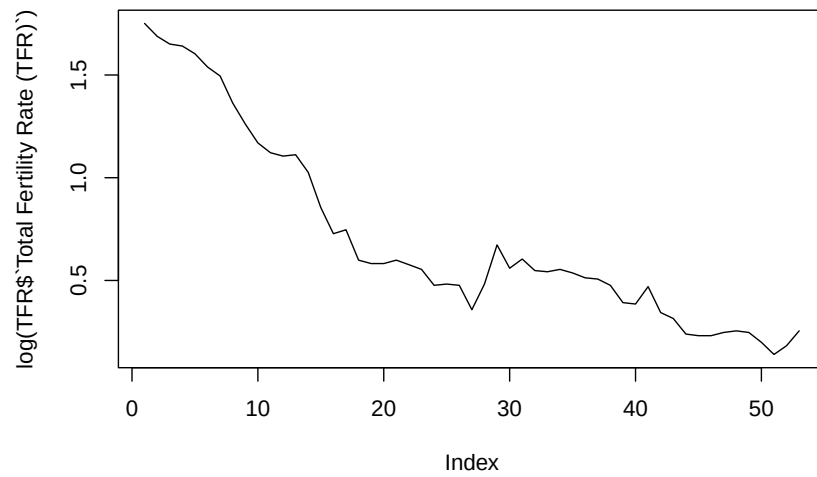
Fitted ARIMA(12,2,3) model for TFR:

$$\begin{aligned}
& (1 + 0.4622B + 0.7606B^2 + 0.9273B^3 + 0.8808B^4 + 0.6967B^5 + 0.4124B^6 \\
& \quad + 0.2312B^7 - 0.0490B^8 - 0.1578B^9 - 0.3268B^{10} - 0.1426B^{11} - 0.6463B^{12})(1 - B)^2 y_t \\
& = (1 - 0.1402B + 0.1411B^2 + 0.8593B^3)\varepsilon_t + c
\end{aligned}$$

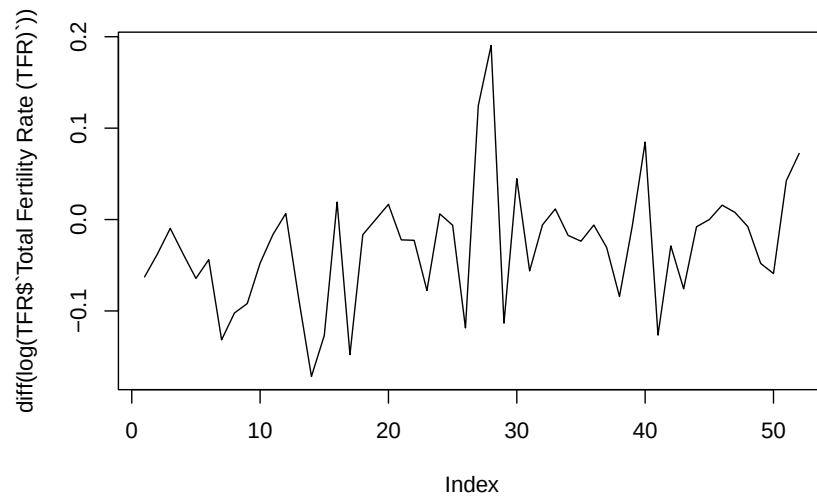
Figure 7. ACF and PACF plots of residuals for the ARIMA(12,2,3) model fitted to TFR



Plot of Log Transformation for Total Fertility Rate (TFR) until 2012

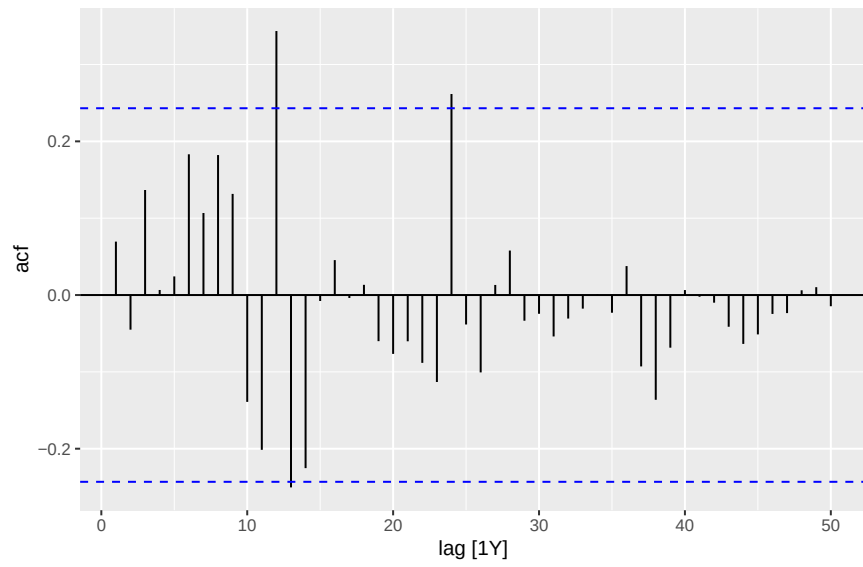


Plot of Differenced Log Transformation for Total Fertility Rate (TFR)

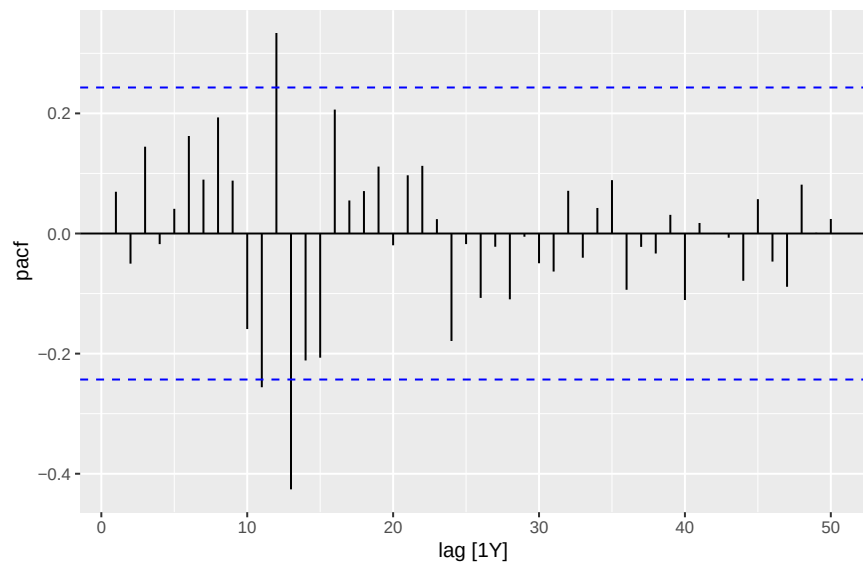


```
##    kpss_stat kpss_pvalue
## 0.50484854 0.04057465
```

ACF Plot for Differenced Log Transformation of Total Fertility Rate (TFR)

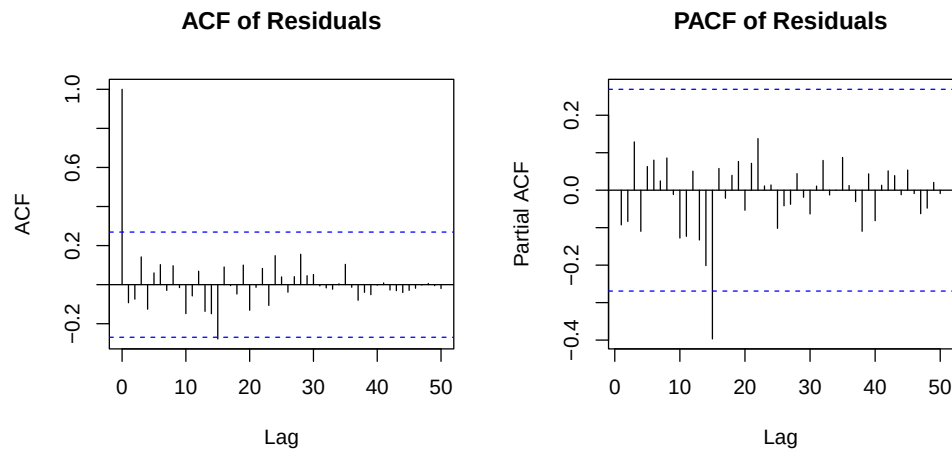


PACF Plot for Differenced Log Transformation of Total Fertility Rate (TFR)



```
## Series: log(TFR_ts)
## ARIMA(13,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2629 -0.0799  0.1194  0.0024  0.1349  0.1715  0.1679  0.2392  0.0298
## s.e.  0.1215  0.1145  0.1065  0.1064  0.1079  0.0974  0.1023  0.1036  0.1106
##      ar10     ar11     ar12     ar13
##      -0.0307 -0.2349  0.3587 -0.4375
## s.e.  0.1130  0.1084  0.1170  0.1258
##
## sigma^2 = 0.00323: log likelihood = 79.9
## AIC=-131.8  AICc=-120.45  BIC=-104.48
```

Figure 8. ACF and PACF plots of residuals for the ARIMA(13,1,0) model fitted to TFR with log transformation



```

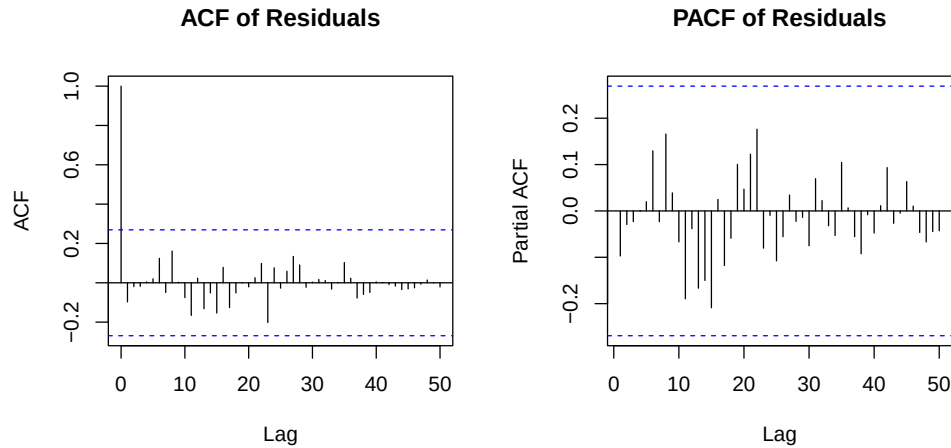
## Series: log(TFR_ts)
## ARIMA(13,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2783 -0.2434 -0.2961 -0.0013  0.1400  0.2218  0.1567  0.2497  0.0839
## s.e.  0.1256  0.1019  0.1096  0.1094  0.1091  0.0975  0.1024  0.1087  0.1147
##      ar10     ar11     ar12     ar13     ma1      ma2      ma3
##      0.0722 -0.0719  0.4674 -0.5235  0.0903  0.1376  0.9749
## s.e.  0.1170  0.1097  0.1005  0.1256  0.0271  0.1085  0.1089
##
## sigma^2 = 0.00232:  log likelihood = 85
## AIC=-136   AICc=-118   BIC=-102.83

```

Fitted ARIMA(13,1,3) model for TFR:

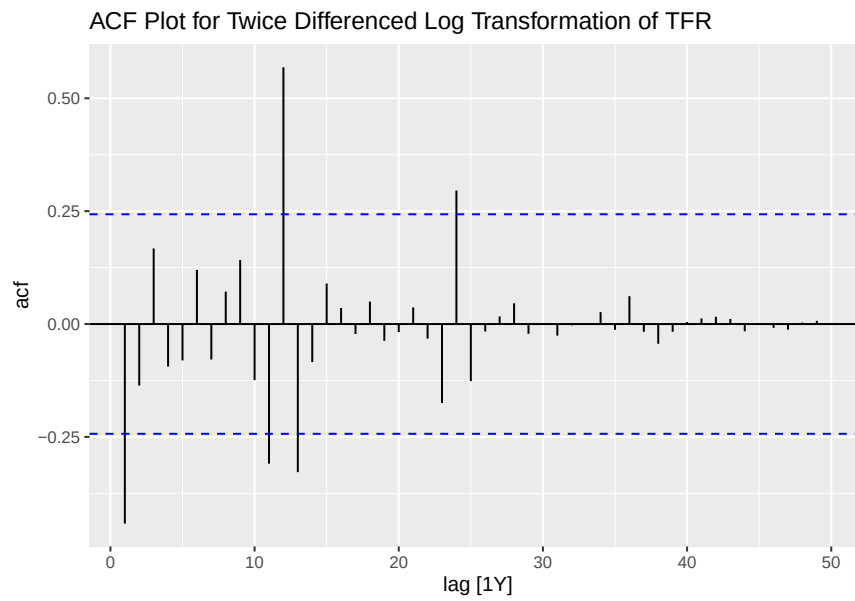
$$\begin{aligned}
& (1 - 0.2783B + 0.2434B^2 + 0.2961B^3 + 0.0013B^4 - 0.1400B^5 - 0.2218B^6 \\
& \quad - 0.1567B^7 - 0.2497B^8 - 0.0839B^9 - 0.0722B^{10} + 0.0719B^{11} - 0.4674B^{12} + 0.5235B^{13})(1 - B) \log(y_t) \\
& = (1 - 0.0903B - 0.1376B^2 - 0.9749B^3)\varepsilon_t + c
\end{aligned}$$

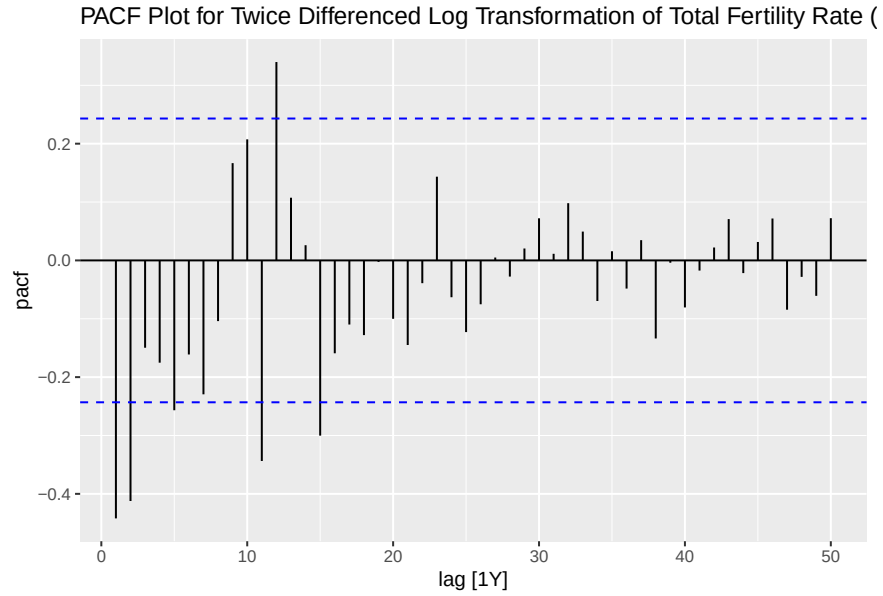
Figure 9. ACF and PACF plots of residuals for the ARIMA(13,1,3) model fitted to TFR with log transformation





```
##      kpss_stat kpss_pvalue
## 0.04760604 0.10000000
```

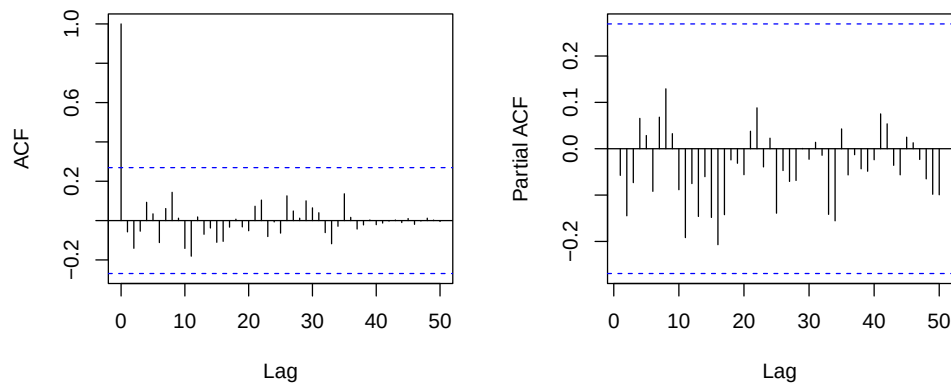




```
## Series: log(TFR_ts)
## ARIMA(15,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      -0.7137 -0.7004 -0.3372 -0.5826 -0.3331 -0.1103  0.0628  0.2087
## s.e.   0.1218  0.1571  0.1913  0.1824  0.2031  0.2111  0.2146  0.2149
##      ar9      ar10     ar11     ar12     ar13     ar14     ar15
##      0.1864  0.1142 -0.1162  0.3260 -0.1661 -0.2596 -0.4876
## s.e.   0.2152  0.2137  0.2042  0.1855  0.1901  0.1618  0.1289
##
## sigma^2 = 0.002602: log likelihood = 82.83
## AIC=-133.67  AICc=-117.67  BIC=-102.76
```

Figure 8. ACF and PACF plots of residuals for the ARIMA(13,1,0) model fitted to TFR with log transformation

Residuals for ARIMA(15,2,0) for TFR with log transformation



```

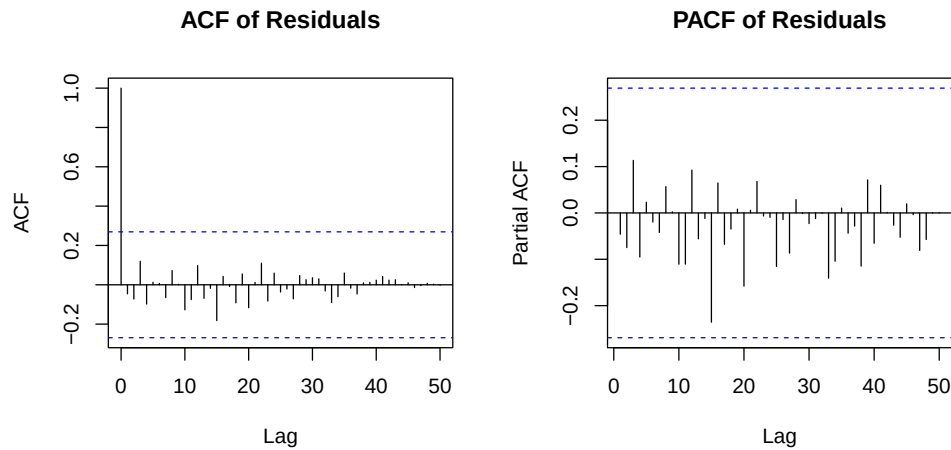
## Series: log(TFR_ts)
## ARIMA(13,2,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6514 -0.2412 -0.0063 -0.1608  0.0605  0.0715  0.0280  0.1159
## s.e.  0.1194  0.1202  0.1193  0.1180  0.1158  0.1095  0.1099  0.1125
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.1105 -0.0649 -0.1966  0.4240 -0.6274 -1.6346  0.7851
## s.e.  0.1156  0.1159  0.1124  0.1126  0.0987  0.1446  0.1392
##
## sigma^2 = 0.002717: log likelihood = 82.19
## AIC=-132.39  AICc=-116.39  BIC=-101.48

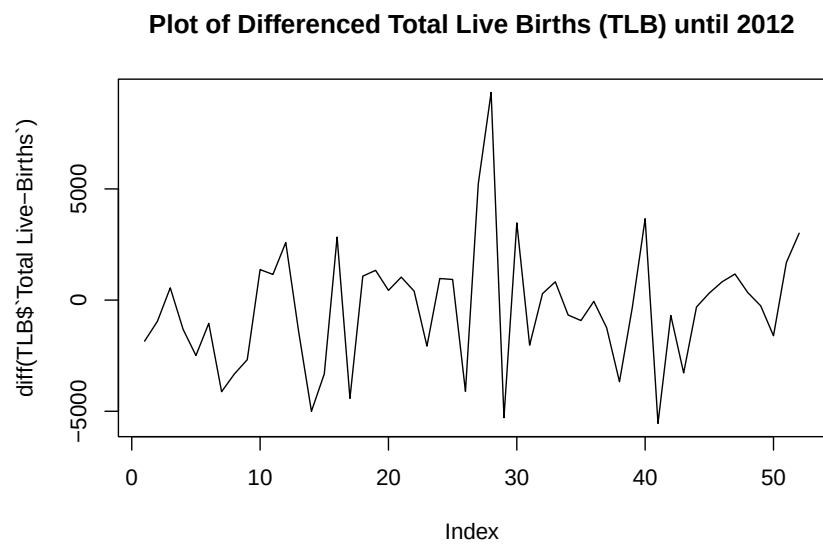
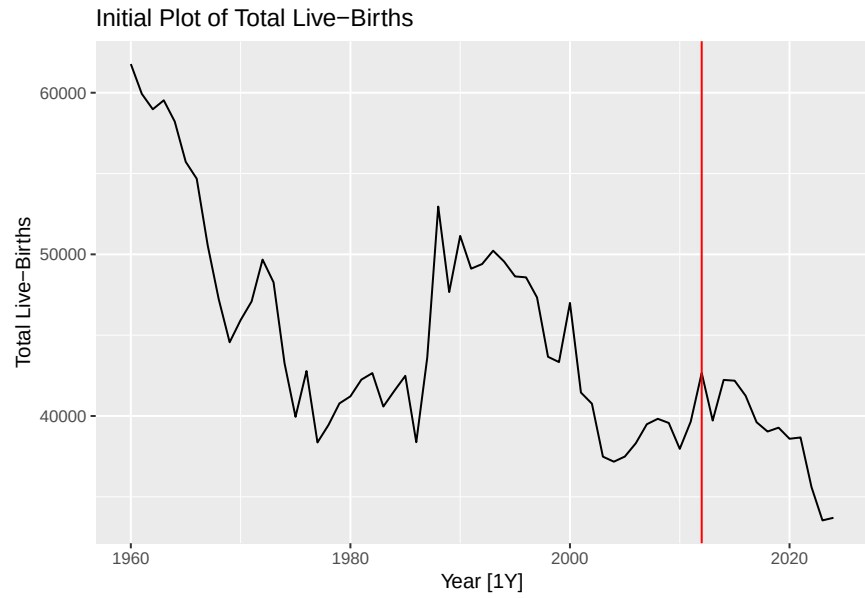
```

Fitted ARIMA(13,2,2) model for TFR:

$$\begin{aligned}
& (1 - 0.6514B + 0.2412B^2 + 0.0063B^3 + 0.1608B^4 - 0.0605B^5 - 0.0715B^6 \\
& \quad - 0.0280B^7 - 0.1159B^8 + 0.1105B^9 + 0.0649B^{10} + 0.1966B^{11} - 0.4240B^{12} + 0.6274B^{13})(1 - B)^2 \log(y_t) \\
& = (1 + 1.6346B - 0.7851B^2)\varepsilon_t + c
\end{aligned}$$

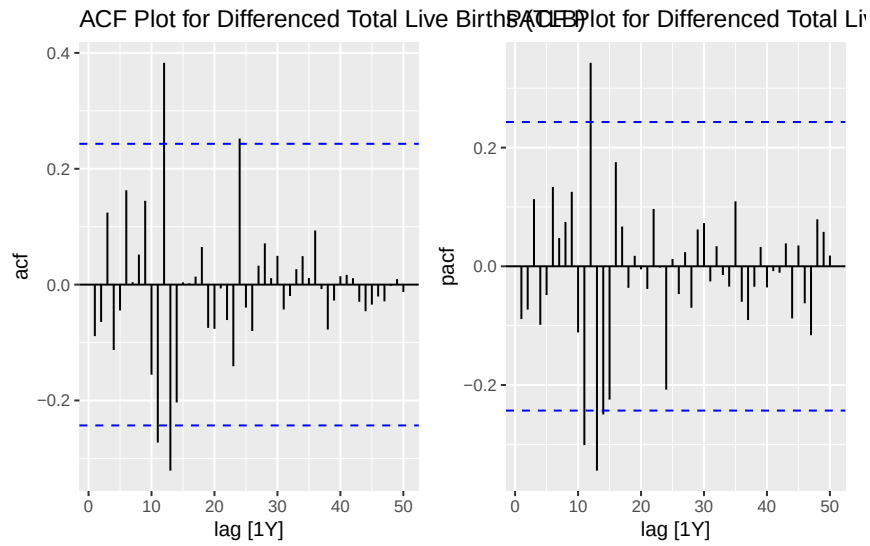
Figure 10. ACF and PACF plots of residuals for the ARIMA(13,2,2) model fitted to TFR with log transformation





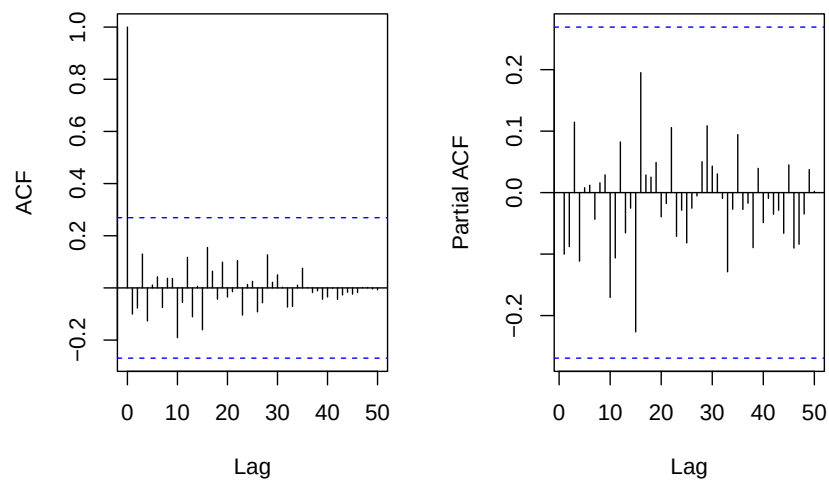
```
##      kpss_stat kpss_pvalue
## 0.2126452    0.1000000
```

Figure 9. ACF and PACF Plots for Differenced Total Live-Births (TLB)



```
## Series: TLB_ts
## ARIMA(14,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.0260  0.0380 -0.0184 -0.0555  0.0794  0.1459  0.1111  0.1666  0.0608
## s.e.  0.1344  0.1218  0.1091  0.1027  0.1050  0.1015  0.1087  0.1051  0.1087
##      ar10     ar11     ar12     ar13     ar14
##      -0.0696 -0.2211  0.2823 -0.3715 -0.2759
## s.e.   0.1085   0.1078  0.1195   0.1245   0.1466
##
## sigma^2 = 5629775:  log likelihood = -472.66
## AIC=975.33   AICc=988.66   BIC=1004.6
```

ACF of Residuals for ARIMA(14,1,0) fACF of Residuals for ARIMA(14,1,0) f

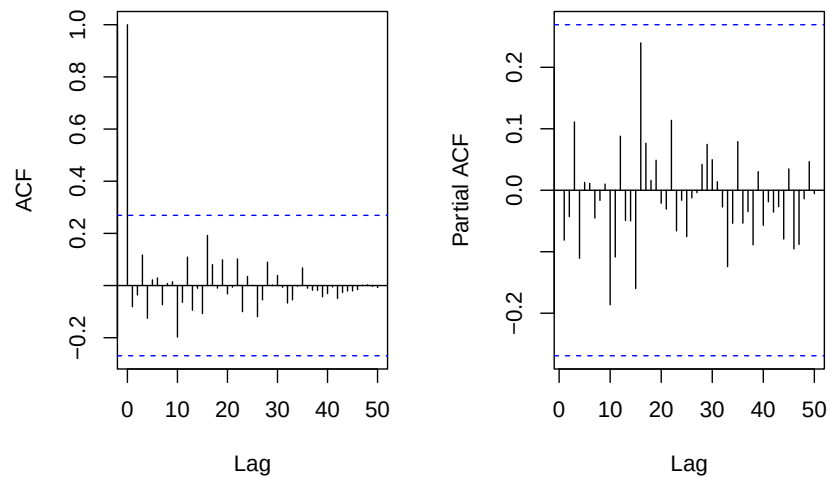


```

## Series: TLB_ts
## ARIMA(13,1,1)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.5571 -0.0703  0.0264 -0.0762  0.0940  0.1004  0.0542  0.1077 -0.0216
## s.e.  0.1677  0.1156  0.1116  0.1120  0.1104  0.1040  0.1090  0.1095  0.1125
##      ar10     ar11     ar12     ar13     ma1
##      -0.0853 -0.1964  0.4091 -0.5605 -0.5344
## s.e.  0.1130  0.1112  0.1223  0.1126  0.1886
##
## sigma^2 = 5503186: log likelihood = -472.28
## AIC=974.57  AICc=987.9  BIC=1003.83

```

CF of Residuals for ARIMA(13,1,1) fCF of Residuals for ARIMA(13,1,1) f

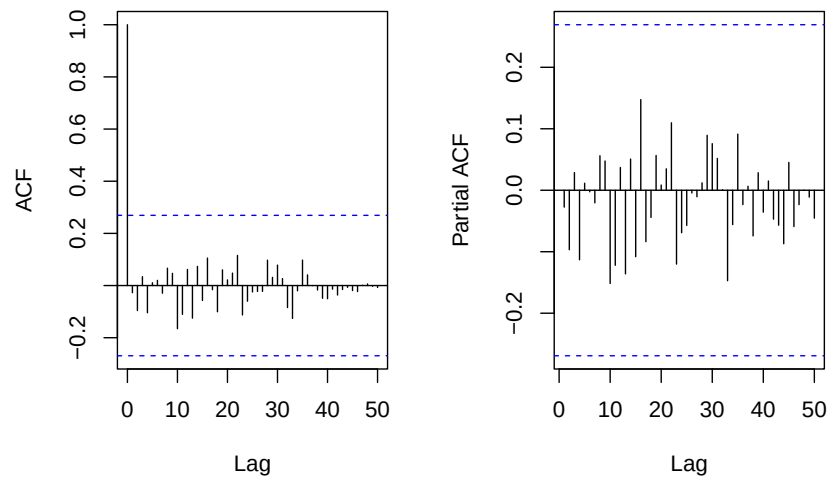


```

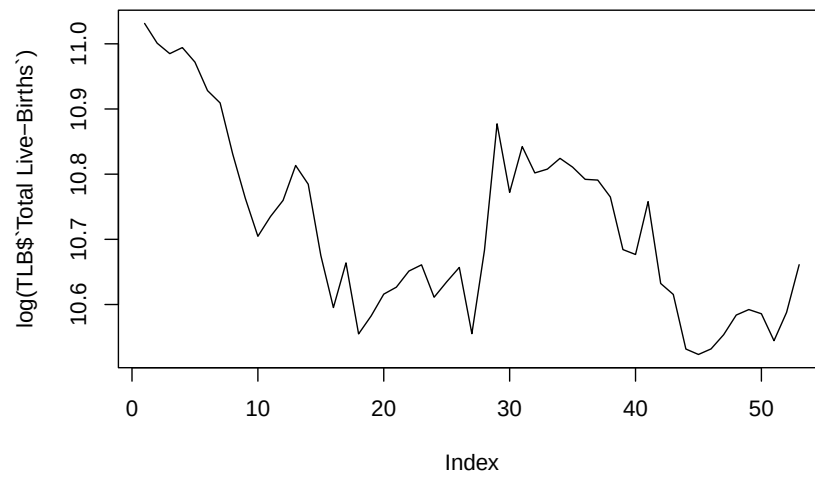
## Series: TLB_ts
## ARIMA(13,1,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6648 -0.4027 -0.0077 -0.0835  0.1408  0.0861  0.0430  0.1468
## s.e.  0.1376  0.1968  0.1260  0.1252  0.1252  0.1203  0.1199  0.1246
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.0053 -0.0478 -0.1305  0.4076 -0.6385 -0.7326  0.5855
## s.e.  0.1262  0.1271  0.1253  0.1159  0.1041  0.1611  0.3171
##
## sigma^2 = 5122169: log likelihood = -470.51
## AIC=973.01  AICc=988.56  BIC=1004.23

```

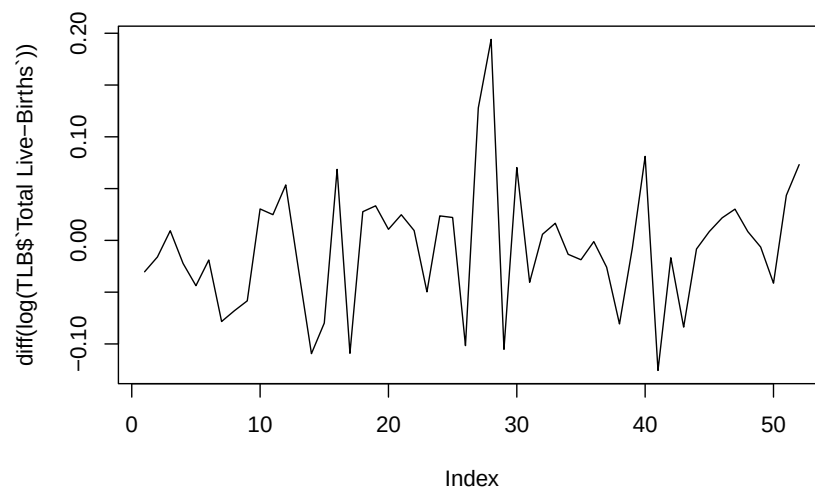
CF of Residuals for ARIMA(13,1,2) fCF of Residuals for ARIMA(13,1,2) f



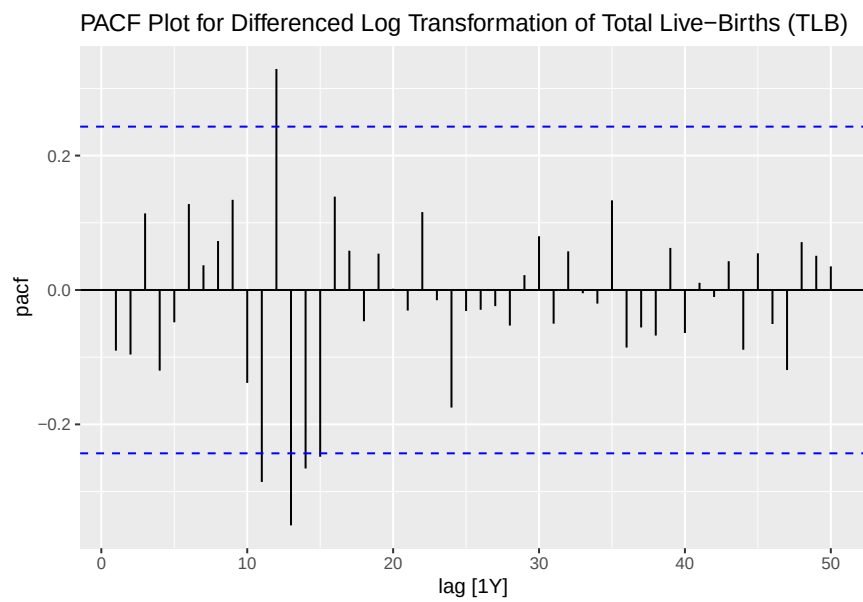
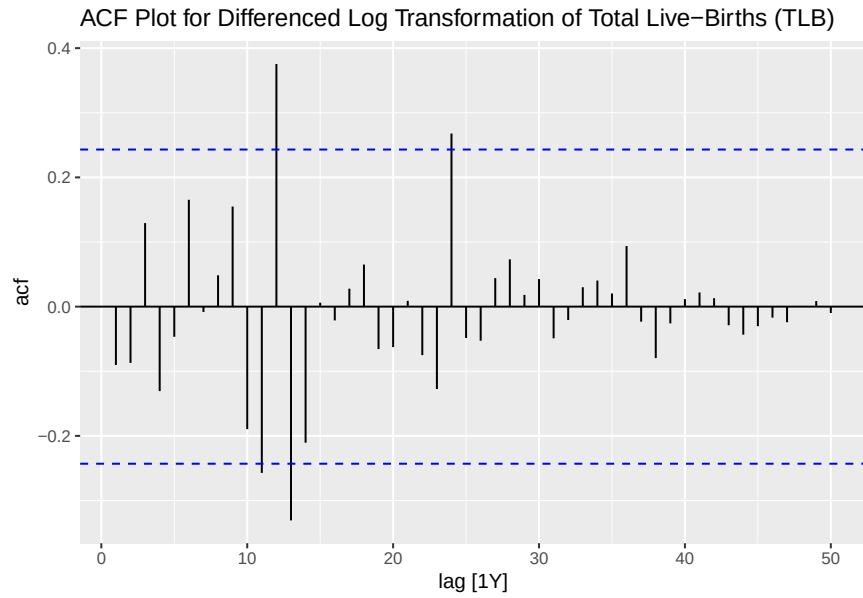
Plot of Log Transformation for Total Live-Births (TLB) until 2012



Plot of Differenced Log Transformation for Total Live-Births (TLB) until

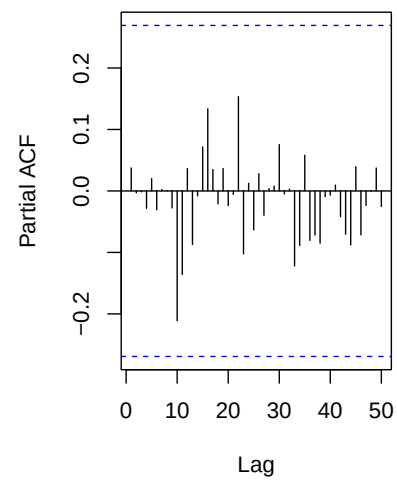
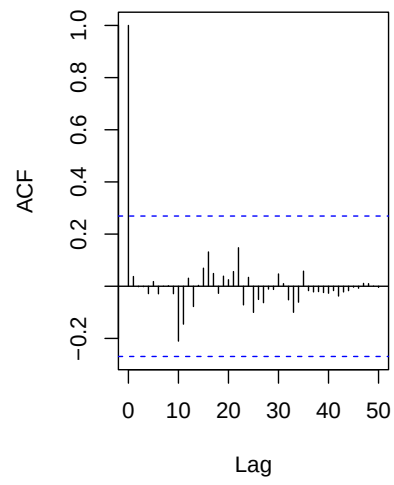


```
##      kpss_stat kpss_pvalue
##      0.1765042  0.1000000
```



```
## Series: log(TLB_ts)
## ARIMA(15,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##    -0.1455 -0.1154  0.0919 -0.1862  0.0220  0.1559  0.1565  0.1742  0.0826
## s.e.   0.1315   0.1197  0.1028  0.0946  0.0935  0.0868  0.0972  0.0918  0.0931
##      ar10     ar11     ar12     ar13     ar14     ar15
##    -0.0706 -0.2325  0.2774 -0.3642 -0.3436 -0.4237
## s.e.   0.0939   0.0892  0.0988  0.1089  0.1264  0.1359
##
## sigma^2 = 0.00225: log likelihood = 88.78
## AIC=-145.56  AICc=-130.02  BIC=-114.34
```


ials for ARIMA(15,1,0) for TLB with uals for ARIMA(15,1,0) for TLB with



```
## Series: log(TLB_ts)
## ARIMA(12,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.3816 -0.3915 -0.4801 -0.1650 -0.0459  0.0622  0.0321  0.1705
## s.e.   0.1314   0.1535   0.1615   0.1495   0.1471   0.1454   0.1494   0.1589
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##      0.1587  0.0805  -0.0368  0.4897  0.5443  0.4755  0.9169
## s.e.   0.1683  0.1646   0.1479  0.1451  0.1011  0.2255  0.2718
##
## sigma^2 = 0.002424: log likelihood = 85.85
## AIC=-139.69  AICc=-124.15  BIC=-108.47
```

Plots for ARIMA(12,1,3) for TLB with Plots for ARIMA(12,1,3) for TLB with

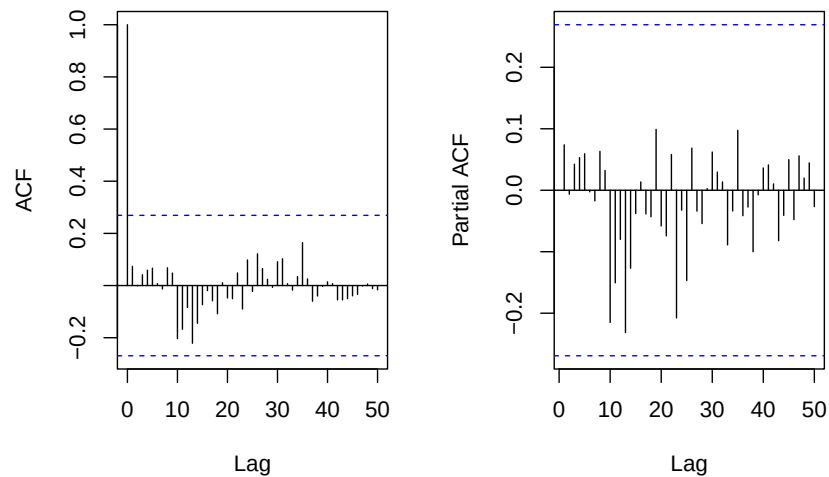


Table 5. Full table for TFR - no log transformation

Model	AIC	AICc	MSE	MAE	
ARIMA(16,1,0)	-63.99	-45.99	0.04753075	0.1690442	Lowest AIC
ARIMA(16,1,1)	-62.39	-42.66	0.04783471	0.1700046	
ARIMA(15,1,1)	-60.26	-42.26	0.08431368	0.223524	Lowest AICc
ARIMA(14,1,3)	-60.99	-40.26	0.01899351	0.09964934	
ARIMA(12,2,3)	-62.57	-46.57	0.01218546	0.08274547	
ARIMA(11,2,3)	-52.91	-39.19	0.05260825	0.1823268	
ARIMA(10,2,3)	-54.69	-43.02	0.05680021	0.1898403	

Table 6. Full table for TFR - log transformed

Model	AIC	AICc	MSE	MAE	
ARIMA(13,1,3)	-136	-118	0.02602064	0.121592	Lowest AIC Log Transform
ARIMA(12,1,4)	-131.83	-113.83	0.02101819	0.1132223	
ARIMA(13,1,5)	-133.96	-110.21	0.05826379	0.1840232	Lowest MSE & MAE
ARIMA(12,1,5)	-130.02	-109.3	0.02172126	0.1147922	
ARIMA(15,2,0)	-133.67	-117.67	0.01869052	0.1056717	
ARIMA(15,2,2)	-135.54	-114.17	0.02235162	0.1187815	
ARIMA(14,2,2)	-130.49	-111.94	0.01087075	0.08352761	
ARIMA(13,2,2)	-132.39	-116.39	0.007426324	0.06826933	
ARIMA(12,2,2)	-127.34	-113.62	0.0498791	0.1720094	

Table 7. Full table for TLB - no log transformation

Model	AIC	AICc	MSE	MAE	
ARIMA(14,1,0)	975.33	988.66	67367429	6853.008	lowest AICc (no log trans- formation)
ARIMA(14,1,1)	975.62	991.17	84179161	7769.997	
ARIMA(13,1,1)	974.57	987.9	82856991	7698.072	
ARIMA(13,1,2)	973.01	988.56	54382525	6014.802	lowest AIC (no log trans- formation), MSE & MAE

Table 8. Full table for TLB - log transformed

Model	AIC	AICc	MSE	MAE	
ARIMA(15,1,0)	-145.56	-130.02	2239721240	47245.34	lowest AICc (log transformed)
ARIMA(15,1,1)	-145.43	-127.43	2173041601	46542.86	
ARIMA(14,1,1)	-140.56	-125.02	2211919788	46958.53	
ARIMA(13,1,1)	-141.45	-128.11	2200303964	46836.59	
ARIMA(12,1,3)	-139.69	-124.15	1752591080	41851.92	

